

Weighting-Like Diagnostics for Nonlinear Bayesian Hierarchical Models

Ryan Giordano, Alice Cima, Erin Hartman, Jared Murray, Avi Feller

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Are US non-voters becoming more Republican?

Blue Rose research says yes:

“Politically disengaged voters have become much more Republican, and because less-engaged voters swung away from [Democrats], an expanded electorate meant a more Republican electorate.”

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(major professional pollsters)

On Data and Democracy says no:

“Claims of a decisive pro-Republican shift among the overall non-voting population are not supported by the most reliable, large-scale post-election data currently available.”

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 - *** **Different statistical methods**
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Our contribution

We define “MrP local equivalent weights” (MrPlew) that:

- Are easily computable from MCMC draws and standard software, and
- Provide MrP versions of key weighting estimator diagnostics.

⇒ **MrPlew provides direct comparisons between MrP and calibration weighting.**

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The key idea is to convert the diagnostic into a *local sensitivity analysis* of this form:

1. Assume your initial model was accurate
2. Select some perturbation your model should be able to capture
3. Use local sensitivity to detect whether the change is what you expect
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I'll do this carefully for covariate balance and MCMC.

But many other variants are possible!

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 - Contrast calibration weighting and MrP
 - Prior work: Equivalent weights for linear models
 - Equivalent weights and implicit weights for non-linear models
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- Locally equivalent weights for frequentist variance estimation
- Locally equivalent weights for covariate balance
 - Describe classical covariate balance
 - Introduce a MrPlew “local empirical consistency check”
 - Theoretical support
 - Examples of real-world results

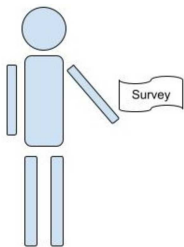
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- Other directions
 - High-level restatement of the logic of our procedure
 - Local versions of other common diagnostics for linear estimators
 - Ongoing and future work

The basic problem

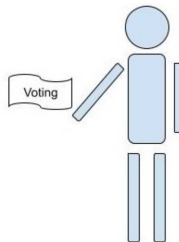
We have a survey population, for whom we observe:

- Covariates $\mathbf{x}$ (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



Observe $(\mathbf{x}_i, y_i)$ for $i = 1, \dots, N_S$



Observe $\mathbf{x}_j$ for $j = 1, \dots, N_T$

¹Photo copyright: Mark Taylor / naturepl.com

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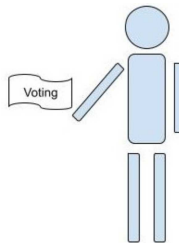
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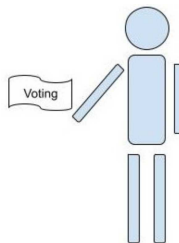
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How can we use the covariates to say something about the target responses?

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Two approaches

We want $\mu := \frac{1}{N_T} \sum_{j=1}^{N_T} y_j$, but don't observe target y_j . Let $Y_S = \{y_1, \dots, y_{N_S}\}$.

- Assume $p(y|\mathbf{x})$ is the same in both populations,
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using only the regressors $\mathbf{x}$
(e.g. raking weights)

Bayesian hierarchical modeling (MrP)

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 - Partial pooling

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 - ← Today, we'll open the box and provide MrP analogues of all these diagnostics